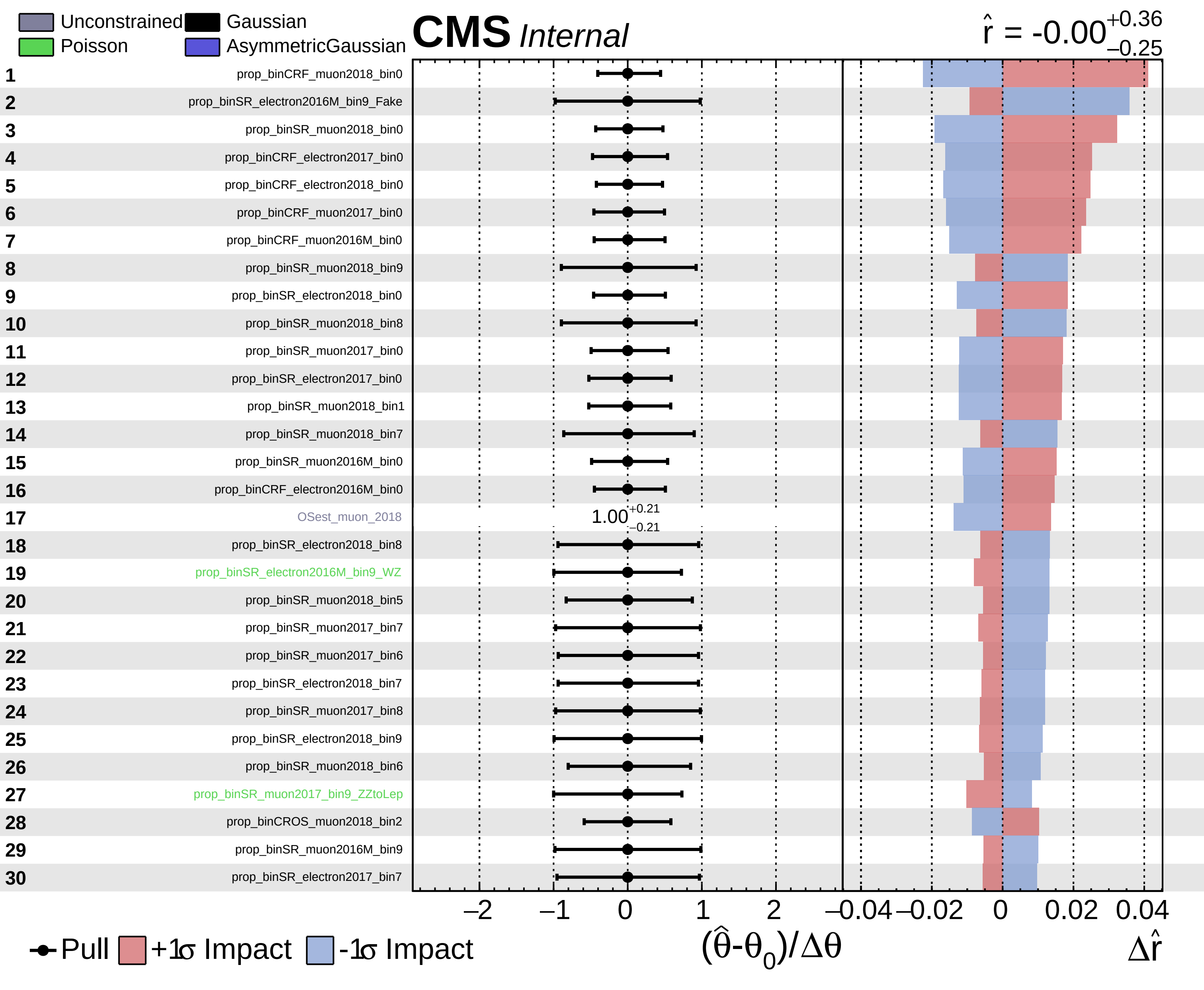
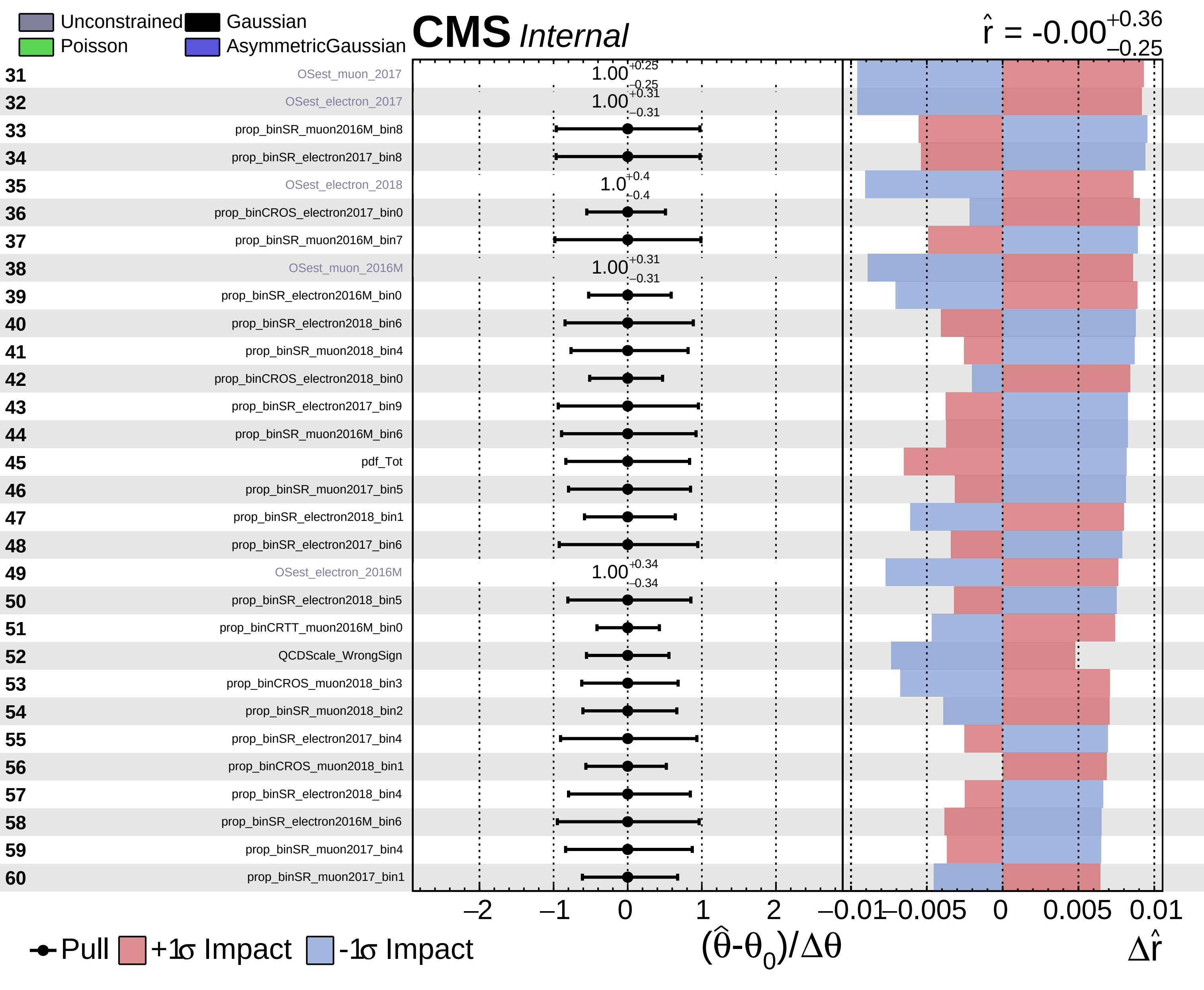
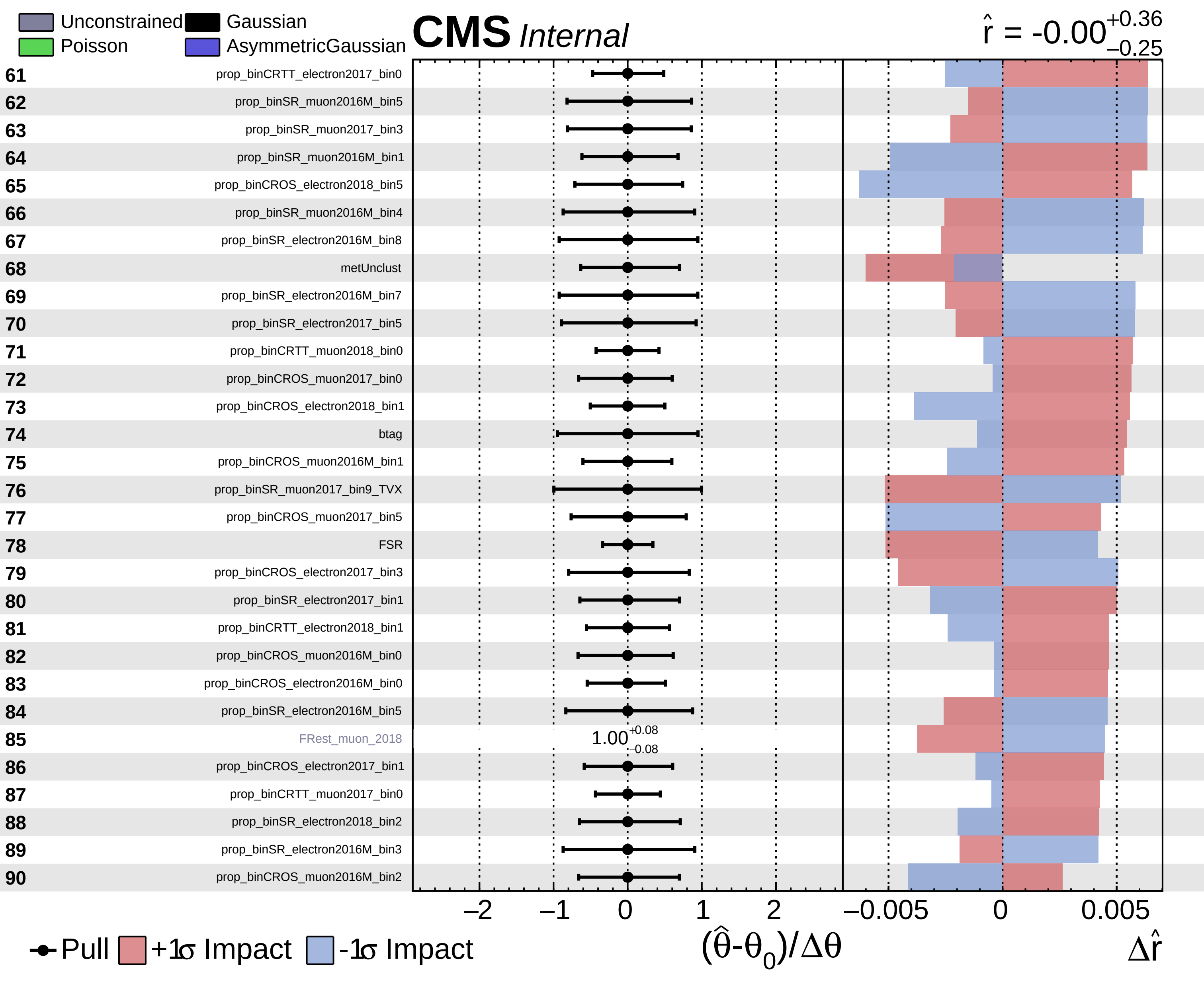
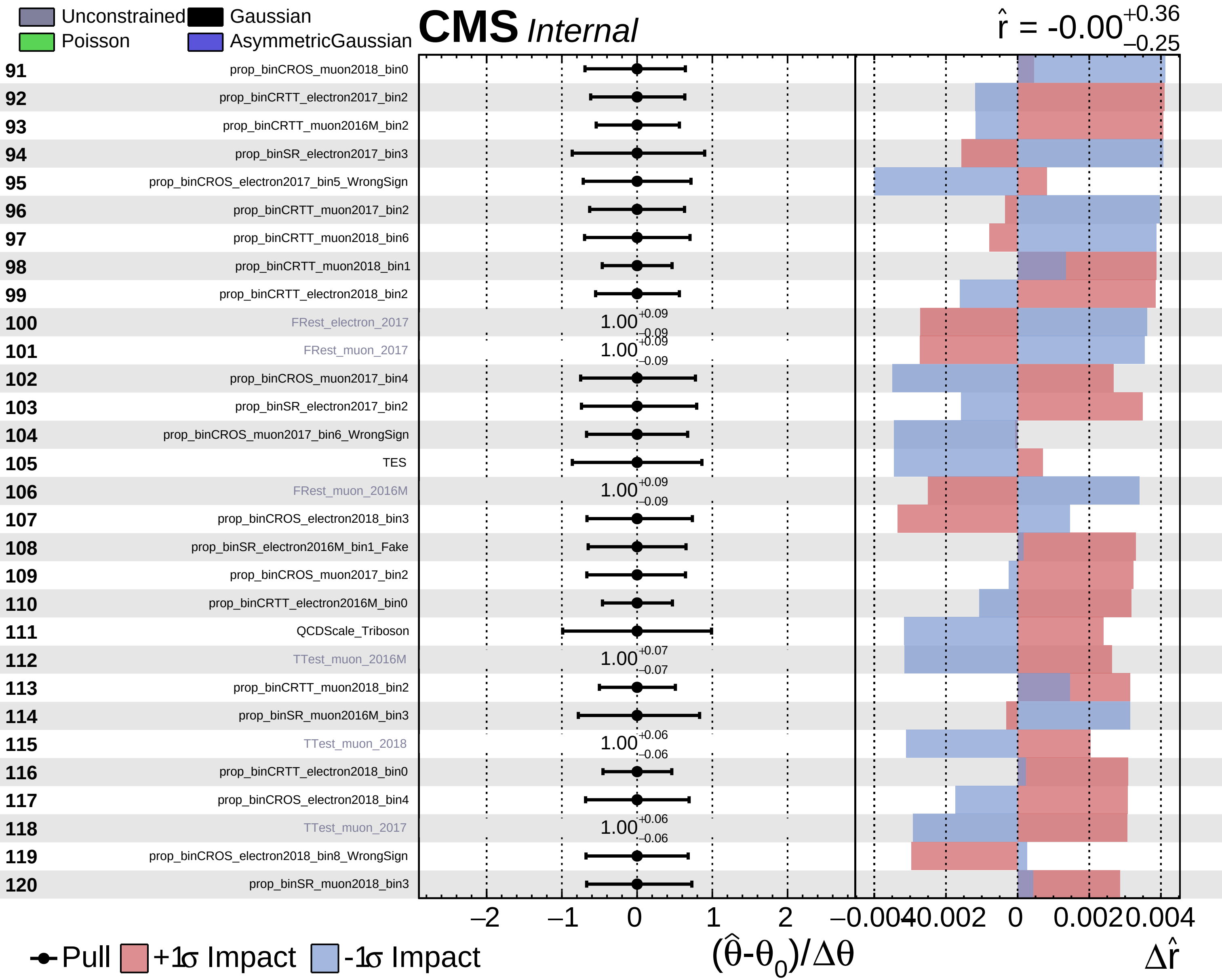


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
1 prop_binCRF_muon2018_bin0
2 prop_binSR_electron2016M_bin9_Fake
3 prop_binSR_muon2018_bin0
4 prop_binCRF_electron2017_bin0
5 prop_binCRF_electron2018_bin0
6 prop_binCRF_muon2017_bin0
7 prop_binCRF_muon2016M_bin0
8 prop_binSR_muon2018_bin9
9 prop_binSR_electron2018_bin0
10 prop_binSR_muon2018_bin8
11 prop_binSR_muon2017_bin0
12 prop_binSR_electron2017_bin0
13 prop_binSR_muon2018_bin1
14 prop_binSR_muon2018_bin7
15 prop_binSR_muon2016M_bin0
16 prop_binCRF_electron2016M_bin0
17 OSest_muon_2018
$1.00^{+0.21}_{-0.21}$
18 prop_binSR_electron2018_bin8
19 prop_binSR_electron2016M_bin9_WZ
20 prop_binSR_muon2018_bin5
21 prop_binSR_muon2017_bin7
22 prop_binSR_muon2017_bin6
23 prop_binSR_electron2018_bin7
24 prop_binSR_muon2017_bin8
25 prop_binSR_electron2018_bin9
26 prop_binSR_muon2018_bin6
27 prop_binSR_muon2017_bin9_ZZtoLep
28 prop_binCROS_muon2018_bin2
29 prop_binSR_muon2016M_bin9
30 prop_binSR_electron2017_bin7
−2
−1
0
1
2
−0.04
−0.02
0
0.02
0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

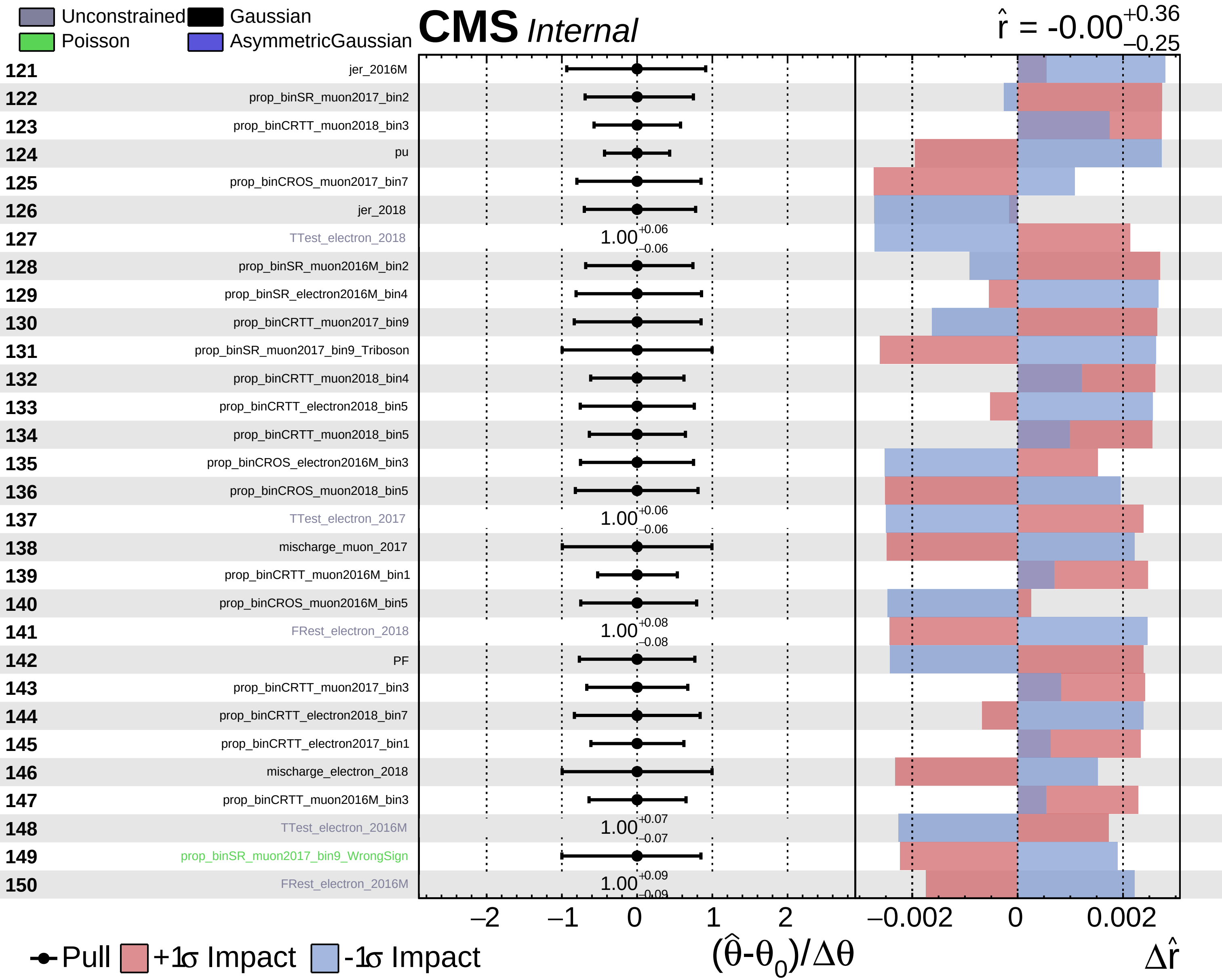


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
31 OSest_muon_2017 $1.00^{+0.25}_{-0.25}$
32 OSest_electron_2017 $1.00^{+0.31}_{-0.31}$
33 prop_binSR_muon2016M_bin8
34 prop_binSR_electron2017_bin8
35 OSest_electron_2018 $1.0^{+0.4}_{-0.4}$
36 prop_binCROS_electron2017_bin0
37 prop_binSR_muon2016M_bin7
38 OSest_muon_2016M $1.00^{+0.31}_{-0.31}$
39 prop_binSR_electron2016M_bin0
40 prop_binSR_electron2018_bin6
41 prop_binSR_muon2018_bin4
42 prop_binCROS_electron2018_bin0
43 prop_binSR_electron2017_bin9
44 prop_binSR_muon2016M_bin6
45 pdf_Tot
46 prop_binSR_muon2017_bin5
47 prop_binSR_electron2018_bin1
48 prop_binSR_electron2017_bin6
49 OSest_electron_2016M $1.00^{+0.34}_{-0.34}$
50 prop_binSR_electron2018_bin5
51 prop_binCRTT_muon2016M_bin0
52 QCDScale_WrongSign
53 prop_binCROS_muon2018_bin3
54 prop_binSR_muon2018_bin2
55 prop_binSR_electron2017_bin4
56 prop_binCROS_muon2018_bin1
57 prop_binSR_electron2018_bin4
58 prop_binSR_electron2016M_bin6
59 prop_binSR_muon2017_bin4
60 prop_binSR_muon2017_bin1
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

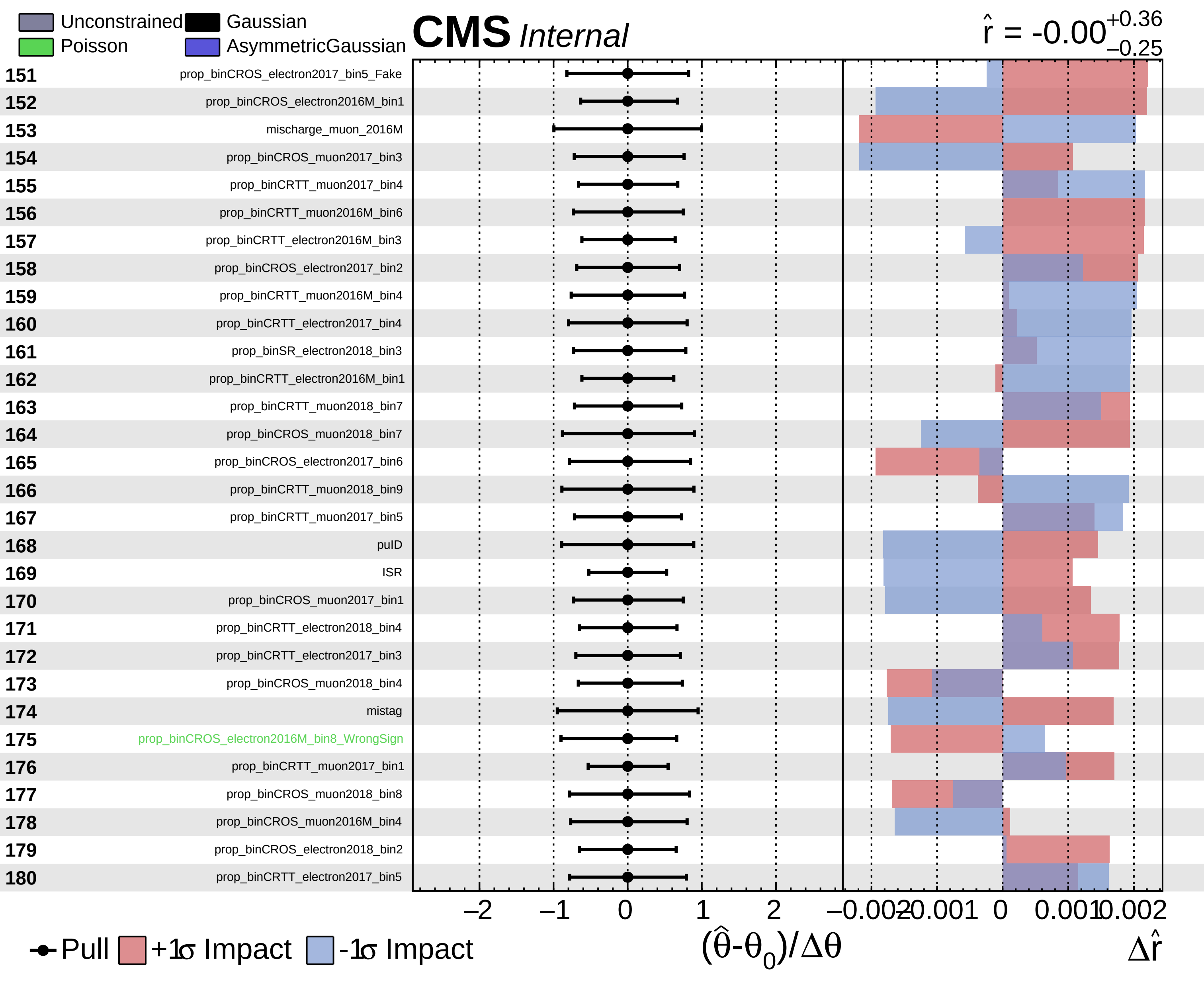

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
61 prop_binCRTT_electron2017_bin0
62 prop_binSR_muon2016M_bin5
63 prop_binSR_muon2017_bin3
64 prop_binSR_muon2016M_bin1
65 prop_binCROS_electron2018_bin5
66 prop_binSR_muon2016M_bin4
67 prop_binSR_electron2016M_bin8
68 metUnclust
69 prop_binSR_electron2016M_bin7
70 prop_binSR_electron2017_bin5
71 prop_binCRTT_muon2018_bin0
72 prop_binCROS_muon2017_bin0
73 prop_binCROS_electron2018_bin1
74 btag
75 prop_binCROS_muon2016M_bin1
76 prop_binSR_muon2017_bin9_TVX
77 prop_binCROS_muon2017_bin5
78 FSR
79 prop_binCROS_electron2017_bin3
80 prop_binSR_electron2017_bin1
81 prop_binCRTT_electron2018_bin1
82 prop_binCROS_muon2016M_bin0
83 prop_binCROS_electron2016M_bin0
84 prop_binSR_electron2016M_bin5
85 FRest_muon_2018
$1.00^{+0.08}_{-0.08}$
86 prop_binCROS_electron2017_bin1
87 prop_binCRTT_muon2017_bin0
88 prop_binSR_electron2018_bin2
89 prop_binSR_electron2016M_bin3
90 prop_binCROS_muon2016M_bin2
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



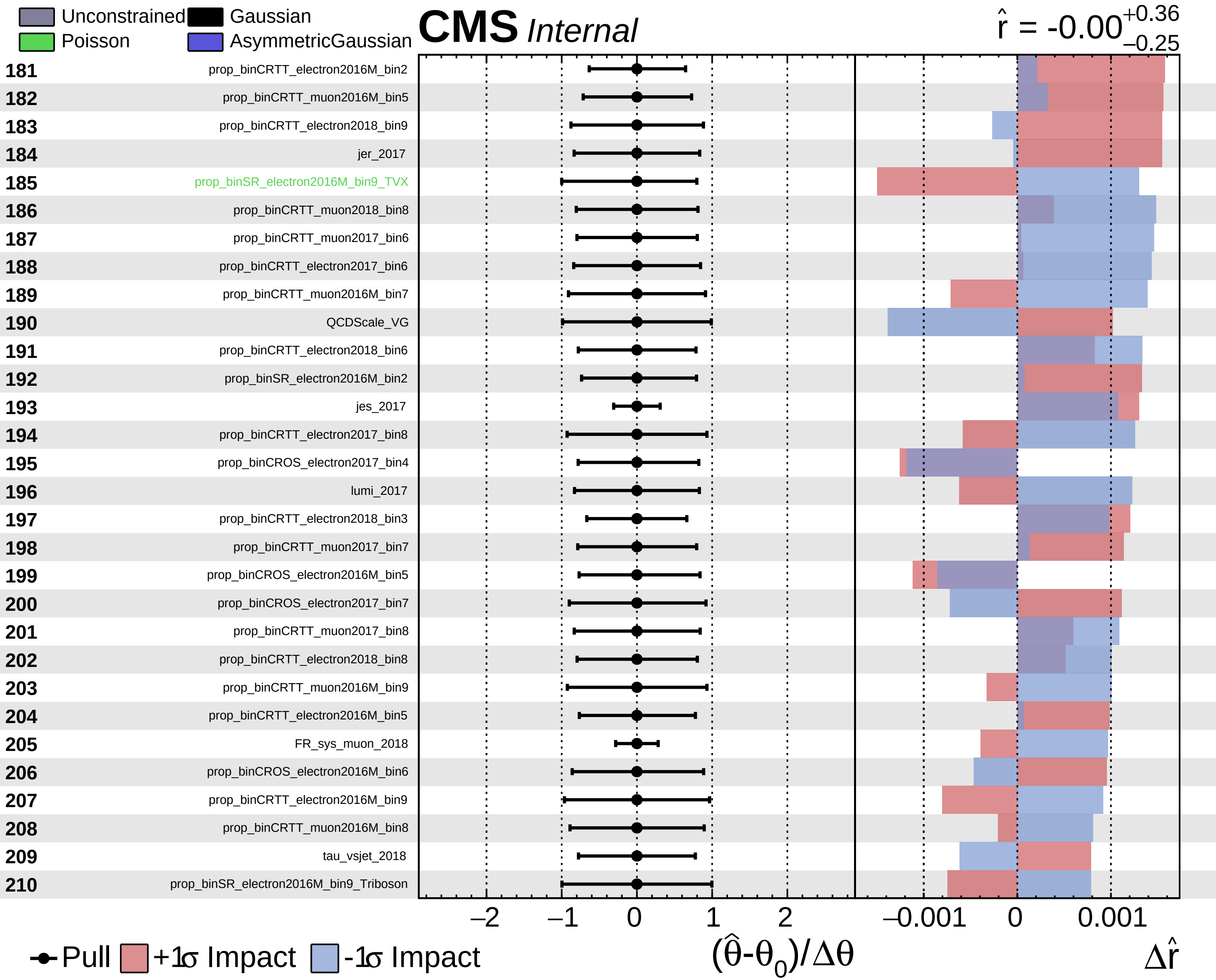
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
91 prop_binCROS_muon2018_bin0
92 prop_binCRTT_electron2017_bin2
93 prop_binCRTT_muon2016M_bin2
94 prop_binSR_electron2017_bin3
95 prop_binCROS_electron2017_bin5_WrongSign
96 prop_binCRTT_muon2017_bin2
97 prop_binCRTT_muon2018_bin6
98 prop_binCRTT_muon2018_bin1
99 prop_binCRTT_electron2018_bin2
100 FRest_electron_2017 $1.00^{+0.09}_{-0.09}$
101 FRest_muon_2017 $1.00^{+0.09}_{-0.09}$
102 prop_binCROS_muon2017_bin4
103 prop_binSR_electron2017_bin2
104 prop_binCROS_muon2017_bin6_WrongSign
105 TES
106 FRest_muon_2016M $1.00^{+0.09}_{-0.09}$
107 prop_binCROS_electron2018_bin3
108 prop_binSR_electron2016M_bin1_Fake
109 prop_binCROS_muon2017_bin2
110 prop_binCRTT_electron2016M_bin0
111 QCDScale_Triboson
112 TTest_muon_2016M $1.00^{+0.07}_{-0.07}$
113 prop_binCRTT_muon2018_bin2
114 prop_binSR_muon2016M_bin3
115 TTest_muon_2018 $1.00^{+0.06}_{-0.06}$
116 prop_binCRTT_electron2018_bin0
117 prop_binCROS_electron2018_bin4
118 TTest_muon_2017 $1.00^{+0.06}_{-0.06}$
119 prop_binCROS_electron2018_bin8_WrongSign
120 prop_binSR_muon2018_bin3
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

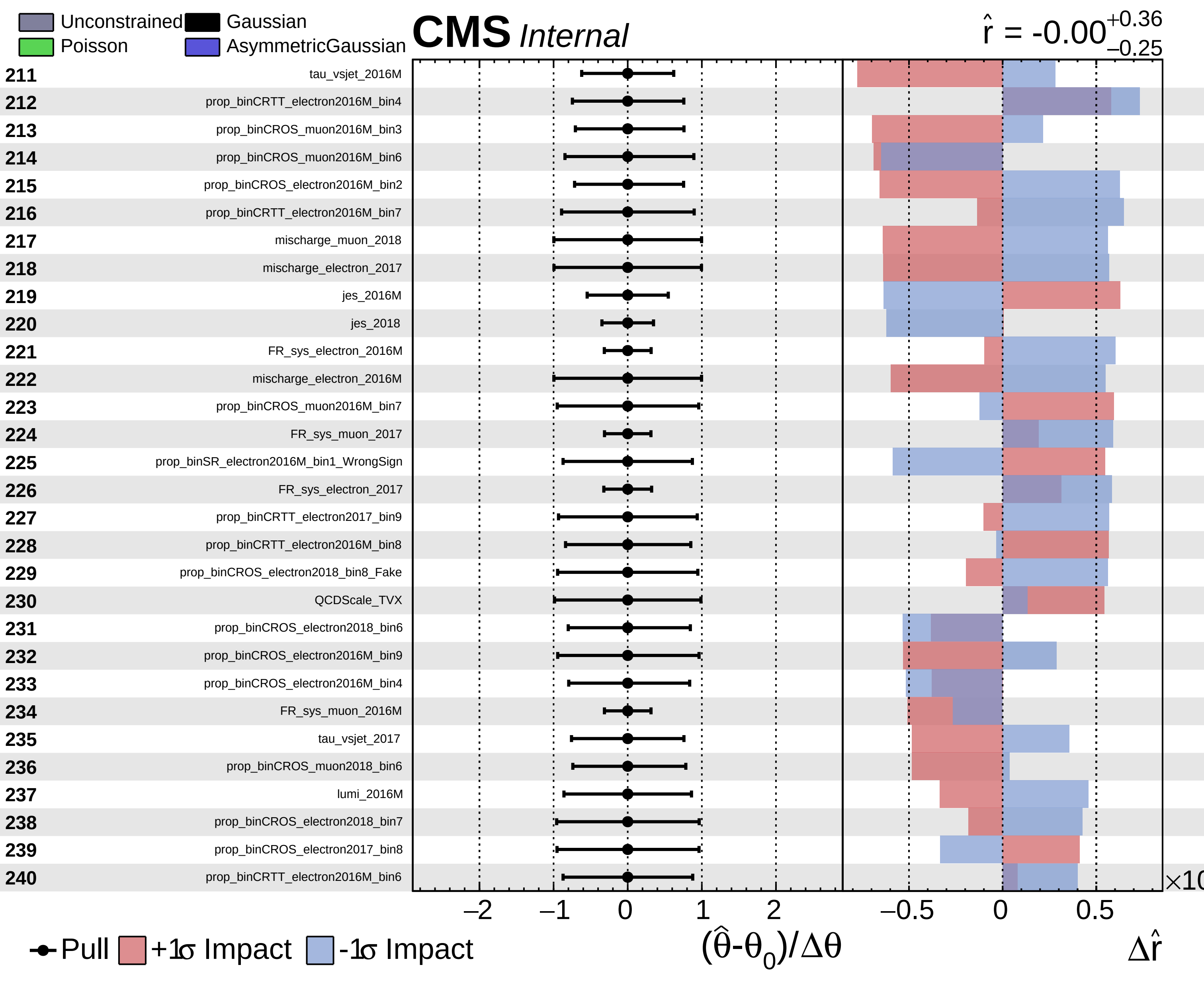


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
121 jer_2016M
122 prop_binSR_muon2017_bin2
123 prop_binCRTT_muon2018_bin3
124 pu
125 prop_binCROS_muon2017_bin7
126 jer_2018
127 TTest_electron_2018 $1.00^{+0.06}_{-0.06}$
128 prop_binSR_muon2016M_bin2
129 prop_binSR_electron2016M_bin4
130 prop_binCRTT_muon2017_bin9
131 prop_binSR_muon2017_bin9_Triboson
132 prop_binCRTT_muon2018_bin4
133 prop_binCRTT_electron2018_bin5
134 prop_binCRTT_muon2018_bin5
135 prop_binCROS_electron2016M_bin3
136 prop_binCROS_muon2018_bin5
137 TTest_electron_2017 $1.00^{+0.06}_{-0.06}$
138 mischarge_muon_2017
139 prop_binCRTT_muon2016M_bin1
140 prop_binCROS_muon2016M_bin5
141 FRest_electron_2018 $1.00^{+0.08}_{-0.08}$
142 PF
143 prop_binCRTT_muon2017_bin3
144 prop_binCRTT_electron2018_bin7
145 prop_binCRTT_electron2017_bin1
146 mischarge_electron_2018
147 prop_binCRTT_muon2016M_bin3
148 TTest_electron_2016M $1.00^{+0.07}_{-0.07}$
149 prop_binSR_muon2017_bin9_WrongSign
150 FRest_electron_2016M $1.00^{+0.09}_{-0.09}$
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

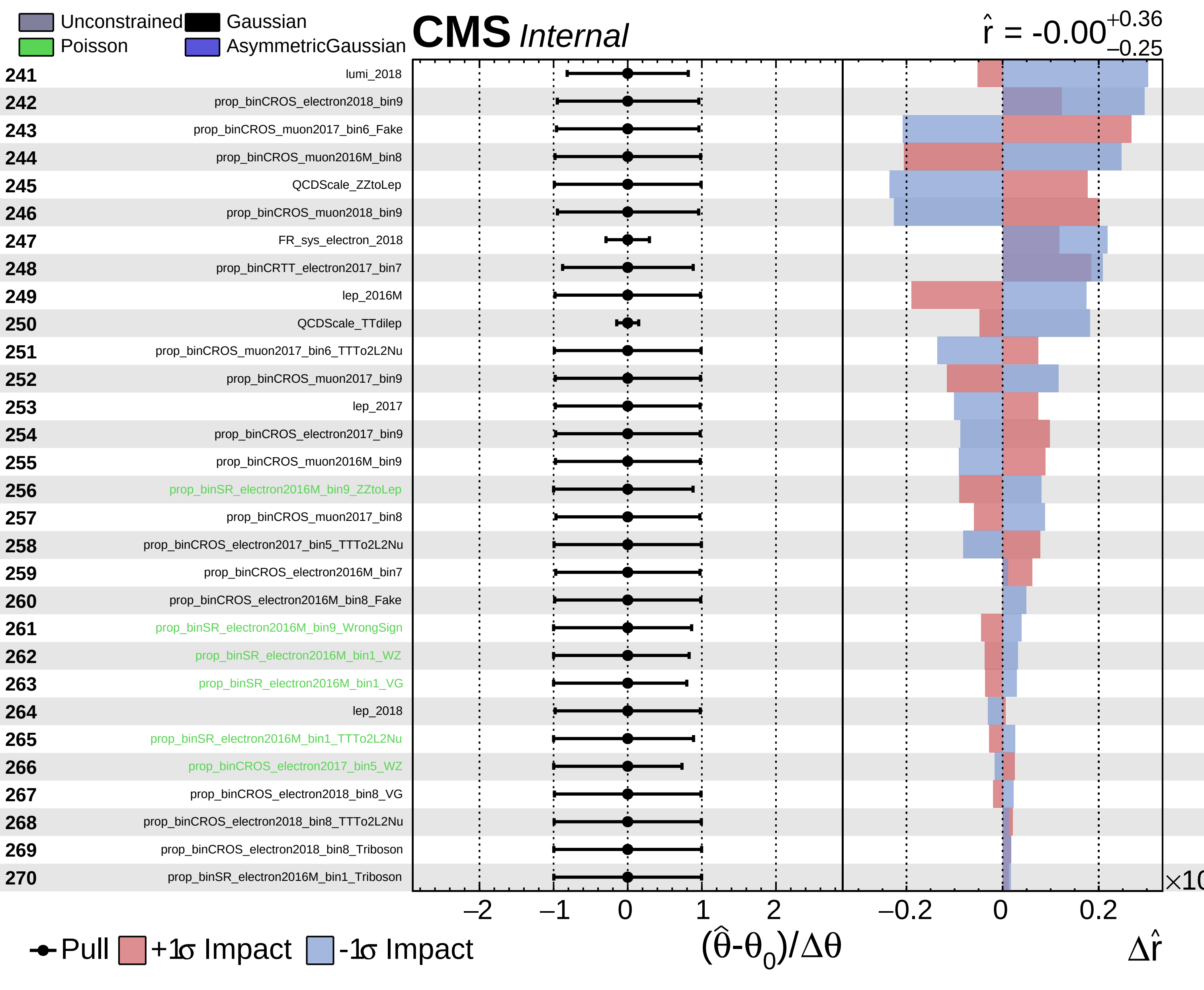


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.36 −0.25
151 prop_binCROS_electron2017_bin5_Fake
152 prop_binCROS_electron2016M_bin1
153 mischarge_muon_2016M
154 prop_binCROS_muon2017_bin3
155 prop_binCRTT_muon2017_bin4
156 prop_binCRTT_muon2016M_bin6
157 prop_binCRTT_electron2016M_bin3
158 prop_binCROS_electron2017_bin2
159 prop_binCRTT_muon2016M_bin4
160 prop_binCRTT_electron2017_bin4
161 prop_binSR_electron2018_bin3
162 prop_binCRTT_electron2016M_bin1
163 prop_binCRTT_muon2018_bin7
164 prop_binCROS_muon2018_bin7
165 prop_binCROS_electron2017_bin6
166 prop_binCRTT_muon2018_bin9
167 prop_binCRTT_muon2017_bin5
168 puID
169 ISR
170 prop_binCROS_muon2017_bin1
171 prop_binCRTT_electron2018_bin4
172 prop_binCRTT_electron2017_bin3
173 prop_binCROS_muon2018_bin4
174 mistag
175 prop_binCROS_electron2016M_bin8_WrongSign
176 prop_binCRTT_muon2017_bin1
177 prop_binCROS_muon2018_bin8
178 prop_binCROS_muon2016M_bin4
179 prop_binCROS_electron2018_bin2
180 prop_binCRTT_electron2017_bin5
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

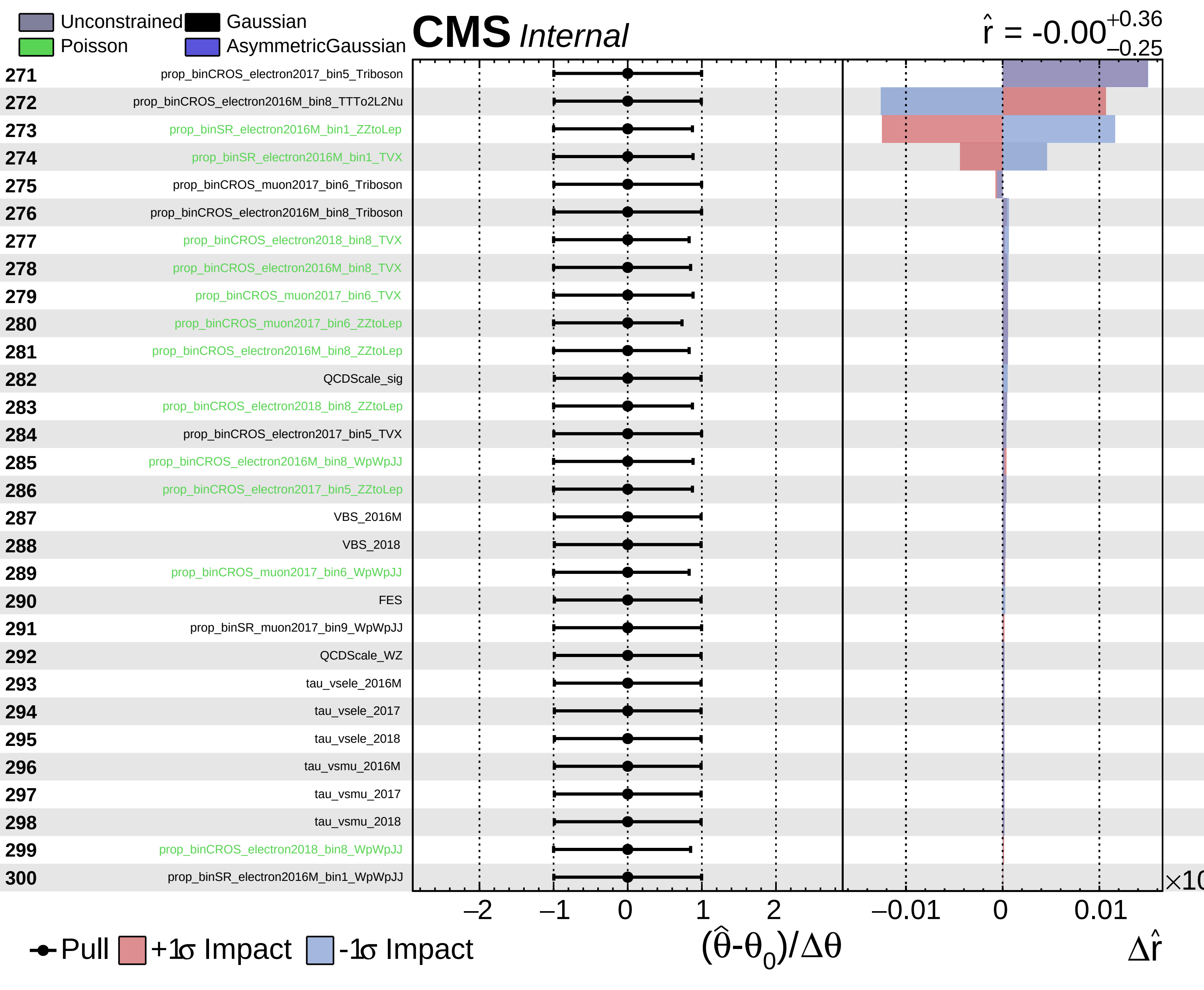

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
181 prop_binCRTT_electron2016M_bin2
182 prop_binCRTT_muon2016M_bin5
183 prop_binCRTT_electron2018_bin9
184 jer_2017
185 prop_binSR_electron2016M_bin9_TVX
186 prop_binCRTT_muon2018_bin8
187 prop_binCRTT_muon2017_bin6
188 prop_binCRTT_electron2017_bin6
189 prop_binCRTT_muon2016M_bin7
190 QCDScale_VG
191 prop_binCRTT_electron2018_bin6
192 prop_binSR_electron2016M_bin2
193 jes_2017
194 prop_binCRTT_electron2017_bin8
195 prop_binCROS_electron2017_bin4
196 lumi_2017
197 prop_binCRTT_electron2018_bin3
198 prop_binCRTT_muon2017_bin7
199 prop_binCROS_electron2016M_bin5
200 prop_binCROS_electron2017_bin7
201 prop_binCRTT_muon2017_bin8
202 prop_binCRTT_electron2018_bin8
203 prop_binCRTT_muon2016M_bin9
204 prop_binCRTT_electron2016M_bin5
205 FR_sys_muon_2018
206 prop_binCROS_electron2016M_bin6
207 prop_binCRTT_electron2016M_bin9
208 prop_binCRTT_muon2016M_bin8
209 tau_vsjet_2018
210 prop_binSR_electron2016M_bin9_Triboson
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



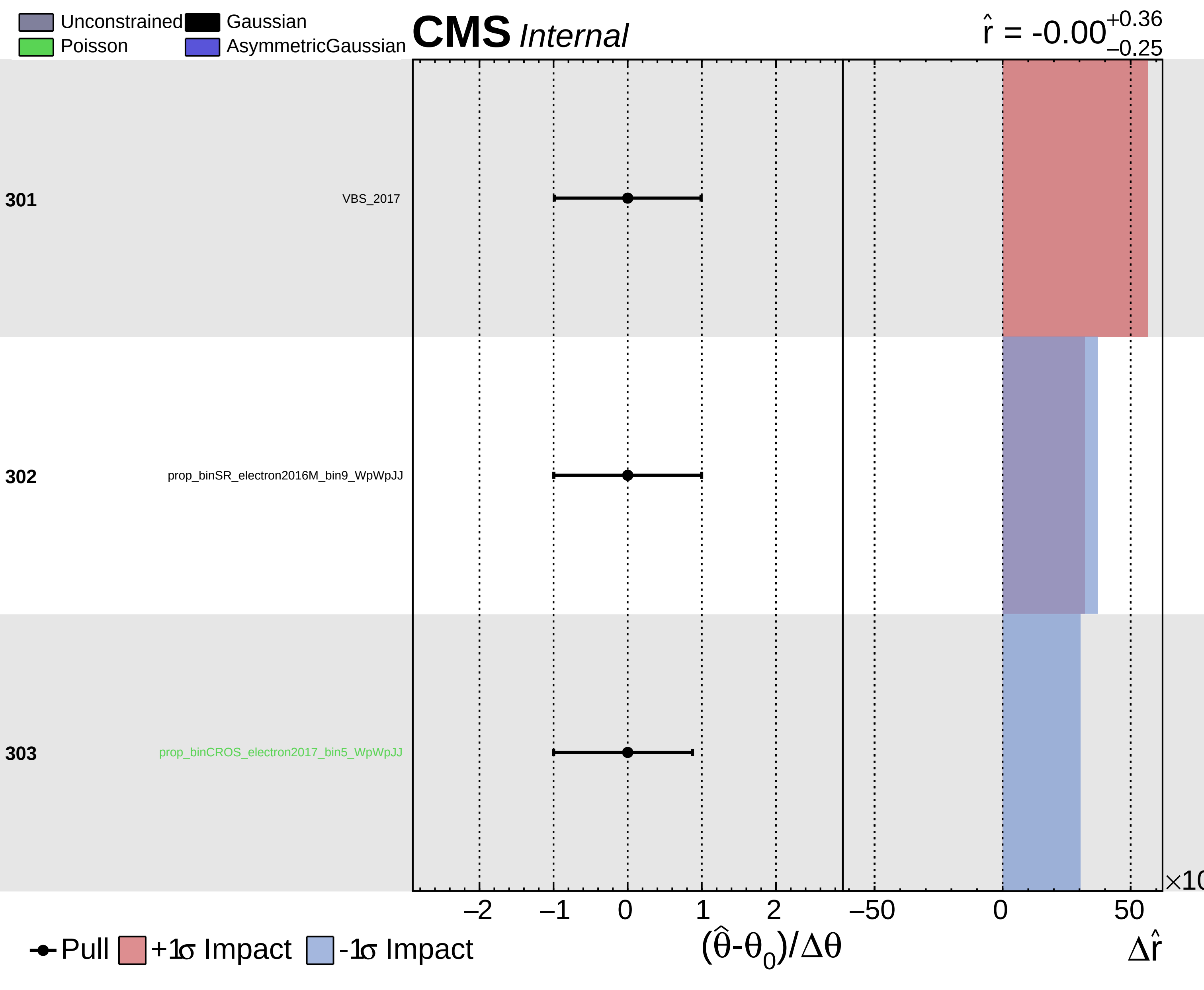
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
211 tau_vsjet_2016M
212 prop_binCRTT_electron2016M_bin4
213 prop_binCROS_muon2016M_bin3
214 prop_binCROS_muon2016M_bin6
215 prop_binCROS_electron2016M_bin2
216 prop_binCRTT_electron2016M_bin7
217 mischarge_muon_2018
218 mischarge_electron_2017
219 jes_2016M
220 jes_2018
221 FR_sys_electron_2016M
222 mischarge_electron_2016M
223 prop_binCROS_muon2016M_bin7
224 FR_sys_muon_2017
225 prop_binSR_electron2016M_bin1_WrongSign
226 FR_sys_electron_2017
227 prop_binCRTT_electron2017_bin9
228 prop_binCRTT_electron2016M_bin8
229 prop_binCROS_electron2018_bin8_Fake
230 QCDScale_TVX
231 prop_binCROS_electron2018_bin6
232 prop_binCROS_electron2016M_bin9
233 prop_binCROS_electron2016M_bin4
234 FR_sys_muon_2016M
235 tau_vsjet_2017
236 prop_binCROS_muon2018_bin6
237 lumi_2016M
238 prop_binCROS_electron2018_bin7
239 prop_binCROS_electron2017_bin8
240 prop_binCRTT_electron2016M_bin6
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.36 −0.25
241 lumi_2018
242 prop_binCROS_electron2018_bin9
243 prop_binCROS_muon2017_bin6_Fake
244 prop_binCROS_muon2016M_bin8
245 QCDScale_ZZtoLep
246 prop_binCROS_muon2018_bin9
247 FR_sys_electron_2018
248 prop_binCRTT_electron2017_bin7
249 lep_2016M
250 QCDScale_TTdilep
251 prop_binCROS_muon2017_bin6_TTTo2L2Nu
252 prop_binCROS_muon2017_bin9
253 lep_2017
254 prop_binCROS_electron2017_bin9
255 prop_binCROS_muon2016M_bin9
256 prop_binSR_electron2016M_bin9_ZZtoLep
257 prop_binCROS_muon2017_bin8
258 prop_binCROS_electron2017_bin5_TTTo2L2Nu
259 prop_binCROS_electron2016M_bin7
260 prop_binCROS_electron2016M_bin8_Fake
261 prop_binSR_electron2016M_bin9_WrongSign
262 prop_binSR_electron2016M_bin1_WZ
263 prop_binSR_electron2016M_bin1_VG
264 lep_2018
265 prop_binSR_electron2016M_bin1_TTTo2L2Nu
266 prop_binCROS_electron2017_bin5_WZ
267 prop_binCROS_electron2018_bin8_VG
268 prop_binCROS_electron2018_bin8_TTTo2L2Nu
269 prop_binCROS_electron2018_bin8_Triboson
270 prop_binSR_electron2016M_bin1_Triboson
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
271 prop_binCROS_electron2017_bin5_Triboson
272 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
273 prop_binSR_electron2016M_bin1_ZZtoLep
274 prop_binSR_electron2016M_bin1_TVX
275 prop_binCROS_muon2017_bin6_Triboson
276 prop_binCROS_electron2016M_bin8_Triboson
277 prop_binCROS_electron2018_bin8_TVX
278 prop_binCROS_electron2016M_bin8_TVX
279 prop_binCROS_muon2017_bin6_TVX
280 prop_binCROS_muon2017_bin6_ZZtoLep
281 prop_binCROS_electron2016M_bin8_ZZtoLep
282 QCDScale_sig
283 prop_binCROS_electron2018_bin8_ZZtoLep
284 prop_binCROS_electron2017_bin5_TVX
285 prop_binCROS_electron2016M_bin8_WpWpJJ
286 prop_binCROS_electron2017_bin5_ZZtoLep
287 VBS_2016M
288 VBS_2018
289 prop_binCROS_muon2017_bin6_WpWpJJ
290 FES
291 prop_binSR_muon2017_bin9_WpWpJJ
292 QCDScale_WZ
293 tau_vsele_2016M
294 tau_vsele_2017
295 tau_vsele_2018
296 tau_vsmu_2016M
297 tau_vsmu_2017
298 tau_vsmu_2018
299 prop_binCROS_electron2018_bin8_WpWpJJ
300 prop_binSR_electron2016M_bin1_WpWpJJ
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.36}_{-0.25}$
301
VBS_2017
302
prop_binSR_electron2016M_bin9_WpWpJJ
303
prop_binCROS_electron2017_bin5_WpWpJJ
−2 −1 0 1 2
−50 0 50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$